

COMPREHENSIVE NUTRITION TEST REPORT

Sample / Illustrative Laboratory Report — Not for clinical diagnosis

Patient	Sample Patient	Report ID	NTR-2026-0908-001
Age / Sex	24 years / Male	Collection Date	08 Sep 2026
Specimen	Serum / Plasma	Report Date	08 Sep 2026
Referring Physician	Preventive Screening	Fasting Status	8 hours

1. Vitamins

Analyte	Result	Unit	Reference Interval	Flag
Vitamin D (25-OH)	18.6	ng/mL	30–100	LOW
Vitamin B12	286	pg/mL	200–900	Within range
Folate (Vitamin B9)	5.8	ng/mL	4.0–20.0	Within range
Vitamin A (Retinol)	42	µg/dL	20–60	Within range
Vitamin E (Alpha-tocopherol)	9.1	mg/L	5.5–17.0	Within range
Vitamin K1	0.78	ng/mL	0.10–2.20	Within range

2. Minerals & Trace Elements

Analyte	Result	Unit	Reference Interval	Flag
Serum Iron	72	µg/dL	60–170	Within range
Ferritin	28	ng/mL	30–400	LOW
Calcium, Total	9.2	mg/dL	8.6–10.2	Within range
Magnesium	1.7	mg/dL	1.7–2.4	Within range
Zinc	64	µg/dL	60–120	Within range
Copper	86	µg/dL	70–140	Within range
Phosphorus	3.4	mg/dL	2.5–4.5	Within range
Selenium	94	µg/L	70–150	Within range

3. Protein & Nutritional Status

Test	Result	Unit	Reference Interval	Flag
Total Protein	7.1	g/dL	6.0–8.3	Within range
Albumin	4.3	g/dL	3.5–5.0	Within range
Globulin	2.8	g/dL	2.0–3.5	Within range
Albumin / Globulin Ratio	1.54	ratio	1.0–2.5	Within range
Prealbumin	22	mg/dL	18–45	Within range

4. Iron Profile

Test	Result	Unit	Reference Interval	Flag
Serum Iron	72	µg/dL	60–170	Within range
TIBC	382	µg/dL	240–450	Within range
Transferrin Saturation	19	%	20–50	LOW
Transferrin	306	mg/dL	200–360	Within range
Ferritin	28	ng/mL	30–400	LOW

5. Metabolic & Lipid Nutrition Markers

Test	Result	Unit	Reference / Desirable	Flag
Fasting Glucose	91	mg/dL	70–99	Within range
HbA1c	5.3	%	<5.7	Within range
Total Cholesterol	178	mg/dL	<200	Desirable
HDL Cholesterol	48	mg/dL	≥40	Within range
LDL Cholesterol	108	mg/dL	<100 optimal	Near optimal
Triglycerides	112	mg/dL	<150	Within range

6. Essential Fatty Acids

Marker	Result	Unit	Interpretive Range	Flag
Omega-3 Index	4.2	%	≥8 desirable	LOW
EPA	0.7	% total fatty acids	0.5–2.0	Within range
DHA	2.1	% total fatty acids	1.0–4.0	Within range
Omega-6 : Omega-3 Ratio	8.4 : 1	ratio	Generally <10 : 1	Within range

7. Electrolytes

Test	Result	Unit	Reference Interval	Flag
Sodium	140	mmol/L	135–145	Within range
Potassium	4.2	mmol/L	3.5–5.1	Within range
Chloride	103	mmol/L	98–107	Within range
Bicarbonate (CO $\blacksquare$)	24	mmol/L	22–29	Within range

8. CBC-Linked Nutritional Indicators

Test	Result	Unit	Reference Interval	Flag
Hemoglobin	13.8	g/dL	13.0–17.0	Within range
MCV	82	fL	80–100	Low-normal
MCH	27.0	pg	27–33	Low-normal
RDW	14.7	%	11.5–14.5	SLIGHTLY HIGH

9. Thyroid-Related Nutritional Marker

Test	Result	Unit	Reference Interval	Flag
TSH	2.18	μIU/mL	0.4–4.0	Within range
Free T4	1.21	ng/dL	0.8–1.8	Within range

10. Overall Nutritional Interpretation

Overall pattern: The sample profile is broadly satisfactory but shows several findings that may warrant dietary review. Vitamin D is below the stated reference interval. Ferritin and transferrin saturation are mildly low, which may be compatible with reduced iron stores and should be interpreted alongside symptoms, dietary intake, and a complete blood count. The Omega-3 Index is below the stated desirable target.

Other markers: Vitamin B12, folate, vitamins A/E/K, calcium, magnesium, zinc, copper, selenium, protein markers, glucose/HbA1c, electrolytes, and most lipid measures are within the illustrative ranges used in this sample report.

Important: This is a fictional sample report created for demonstration/testing purposes. Reference intervals vary by laboratory, assay, age, sex, clinical context, and methodology. Abnormal results should not be used to diagnose a deficiency or disease without clinical assessment.

11. Suggested Nutrition-Focused Follow-Up

Finding	General nutrition consideration	Possible follow-up
Low Vitamin D	Review dietary sources and safe sunlight exposure.	Discuss repeat testing / supplementation with clinician if appropriate.
Low ferritin / saturation	Review iron-rich foods and overall dietary iron intake; pair plant iron with vitamin C.	Consider clinical evaluation of iron status if persistent.
Low Omega-3 Index	Increase omega-3-rich foods such as flax/chia/walnuts or appropriate fish sources.	Reassess based on diet and clinician advice.
Low-normal MCV/MCH	Ensure adequate iron, B12, folate and overall nutrient intake.	Correlate with CBC and iron studies if clinically indicated.

Laboratory note: Results marked LOW/HIGH are illustrative. This document is not a real medical record and contains no real patient data.

Reviewed by	Laboratory Nutrition Panel	Report Status	FINAL — SAMPLE
Signature	_____	Document	Demonstration copy